Blockchain Technology (CSE4444)
Assignment 2

1. Design and develop an original decentralized application (DApp) for a suitable real-world problem using Solidity and the Ethereum blockchain. The application must contain a smart contract, a simple user interface, and MetaMask integration.
   The smart contract must be compiled, deployed, and tested using Remix IDE with Ganache or an Ethereum test network. Students must choose their own application; direct reproduction of examples demonstrated in class is not permitted.
   Minimum Requirements:
   The DApp must include:
   i. A Solidity smart contract with suitable variables, functions, a constructor, and validation.
   ii. A simple frontend using HTML, CSS, and JavaScript.
   iii. MetaMask connection using Web3.js or Ethers.js.
   iv. Successful execution of transactions and retrieval of blockchain data.
   v. Testing with valid, invalid, and unauthorised inputs.

Submission Procedure

Step 1: Complete the Assignment

- Design and implement **one original DApp** that addresses a suitable real-world problem.
- Ensure your submission includes all the required components:
  - Solidity smart contract source file containing suitable variables, functions, a constructor, and validation
  - Frontend files developed using HTML, CSS, and JavaScript
  - MetaMask integration implemented using Web3.js or Ethers.js
  - Successful execution of blockchain transactions
  - Retrieval and display of blockchain data
  - Testing with valid, invalid, and unauthorised inputs
- Compile, deploy, and test the smart contract using **Remix IDE** with **Ganache** or an **Ethereum test network**.

Step 2: Upload Your Work to GitHub

1. Create a **new GitHub repository** with a clear and relevant name. If the repository is private, provide access to the evaluator.
2. Upload **all relevant files and supporting evidence** to the repository, including:
   - Solidity smart contract source file
   - HTML, CSS, and JavaScript frontend files
   - Deployment details
   - Test files or test results

- Transaction evidence and screenshots

3. Include a **README.md** file containing:
    - A description of the problem addressed by the DApp and the steps required to execute the application
    - Steps to connect MetaMask and interact with the smart contract
    - The deployment network and deployed smart contract address
    - Test cases for valid, invalid, and unauthorised inputs, along with the observed results
    - Screenshots showing:
        - Smart contract compilation
        - Smart contract deployment
        - MetaMask wallet connection
        - Successful transaction
        - Failed transaction
        - Blockchain data displayed through the user interface
4. Organize the repository properly and ensure that all submitted files are accessible.
5. **Do not upload** private keys, wallet seed phrases, passwords, or other credentials.

Step 3: Share GitHub Repository Link

1. Create a **Word document** named: YourName_RegNum_BT_Assignment2.docx
2. In the Word document, include:
    - Student name
    - Registration number
    - Assignment title
    - Title of the developed DApp
    - GitHub repository link

Step 4: Convert to PDF

- Convert the Word document to **PDF format**. (File → Save As → PDF)

Step 5: Upload to LMS

- Upload the final **PDF file** to the **Learning Management System (LMS)** under the specified assignment submission section before the deadline.